Conservation Genetics Practice Problems #4

1) Are the models of allele frequency change we discussed in class deterministic or stochastic?

2) Recall that the change in frequency (q) of a recessive allele A_2 under complete dominance and negative selection is $\Delta q = \frac{-spq^2}{1-sq}$. Which coefficient values will lead to greatest single-generation decrease in A_2?

   a) s=0.1, q=0.5, p=0.5
   b) s=0.1, q=0.1, p=0.9
   c) s=0.01, q=0.1, p=0.9
   d) s=0.01, q=0.01, p=0.99

3) Suppose the frequency of a deleterious allele q under incomplete dominance associated with increased susceptibility of blister rust in Whitebark Pine drops from 0.1 to 0.07 from generation $t = 1$ to $t = 2$. Solve for Δp.

4) What mode of inheritence is modeled by $h = 0.7$?

5) Under complete dominance, what selection coefficient indicates the recessive homozygote has a relative fitness of $w(aa) = 0.98$?